- If the even operator has a degenerate spectrum, its eigenvectors do not necessarily have a definite parity.

What about the parity of the position and momentum operators, $\hat{\vec{R}}$ and $\hat{\vec{P}}$? We can easily show that both of them are odd, since they anticommute with the parity operator:

$$\hat{\mathcal{P}}\hat{\vec{R}} = -\hat{\vec{R}}\hat{\mathcal{P}}, \qquad \hat{\mathcal{P}}\hat{\vec{P}} = -\hat{\vec{P}}\hat{\mathcal{P}}; \tag{2.344}$$

hence

$$\hat{\mathcal{P}}\hat{\vec{R}}\hat{\mathcal{P}}^\dagger = -\hat{\vec{R}}, \qquad \hat{\mathcal{P}}\hat{\vec{P}}\hat{\mathcal{P}}^\dagger = -\hat{\vec{P}}, \tag{2.345}$$

since $\hat{\mathcal{P}}\hat{\mathcal{P}}^\dagger = 1$. For instance, to show that $\hat{\vec{R}}$ anticommutes with $\hat{\mathcal{P}}$, we need simply to look at the following relations:

$$\hat{\mathcal{P}}\hat{\vec{R}} \mid \vec{r}\rangle = \vec{r}\hat{\mathcal{P}} \mid \vec{r}\rangle = \vec{r} \mid -\vec{r}\rangle, \tag{2.346}$$

$$\hat{\vec{R}}\hat{\mathcal{P}} \mid \vec{r}\rangle = \hat{\vec{R}} \mid -\vec{r}\rangle = -\vec{r} \mid -\vec{r}\rangle. \tag{2.347}$$

If the operators $\hat{A}$ and $\hat{B}$ are even and odd, respectively, we can verify that

$$\hat{\mathcal{P}}\hat{A}^n\hat{\mathcal{P}} = \hat{A}^n, \qquad \hat{\mathcal{P}}\hat{B}^n\hat{\mathcal{P}} = (-1)^n\hat{B}^n. \tag{2.348}$$

These relations can be shown as follows:

$$\hat{\mathcal{P}}\hat{A}^n\hat{\mathcal{P}} = (\hat{\mathcal{P}}\hat{A}\hat{\mathcal{P}})\,(\hat{\mathcal{P}}\hat{A}\hat{\mathcal{P}})\cdots(\hat{\mathcal{P}}\hat{A}\hat{\mathcal{P}}) = \hat{A}^n, \tag{2.349}$$

$$\hat{\mathcal{P}}\hat{B}^n\hat{\mathcal{P}} = (\hat{\mathcal{P}}\hat{B}\hat{\mathcal{P}})\,(\hat{\mathcal{P}}\hat{B}\hat{\mathcal{P}})\cdots(\hat{\mathcal{P}}\hat{B}\hat{\mathcal{P}}) = (-1)^n\hat{B}^n. \tag{2.350}$$

2.7 Matrix and Wave Mechanics

In this chapter we have so far worked out the mathematics pertaining to quantum mechanics in two different representations: *discrete* basis systems and *continuous* basis systems. The theory of quantum mechanics deals in essence with solving the following eigenvalue problem:

$$\hat{H} \mid \psi\rangle = E \mid \psi\rangle, \tag{2.351}$$

where $\hat{H}$ is the Hamiltonian of the system. This equation is general and does not depend on any coordinate system or representation. But to solve it, we need to represent it in a given basis system. The complexity associated with solving this eigenvalue equation will then vary from one basis to another.

In what follows we are going to examine the representation of this eigenvalue equation in a *discrete* basis and then in a *continuous* basis.

2.7.1 Matrix Mechanics

The representation of quantum mechanics in a *discrete* basis yields a *matrix* eigenvalue problem. That is, the representation of (2.351) in a discrete basis $\{\mid \phi_n\rangle\}$ yields the following matrix

eigenvalue equation (see (2.257)):

$$\begin{vmatrix} H_{11}-E & H_{12} & H_{13} & \cdots & H_{1N} \\ H_{21} & H_{22}-E & H_{23} & \cdots & H_{2N} \\ H_{31} & H_{32} & H_{33}-E & \cdots & H_{3N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ H_{N1} & H_{N2} & H_{N3} & \cdots & H_{NN}-E \end{vmatrix} = 0. \tag{2.352}$$

This is an Nth order equation in E; its solutions yield the energy spectrum of the system: E_1, E_2, E_3, ..., E_N. Knowing the set of eigenvalues E_1, E_2, E_3, ..., E_N, we can easily determine the corresponding set of eigenvectors $|\,\phi_1\rangle$, $|\,\phi_2\rangle$, ..., $|\,\phi_N\rangle$.

The diagonalization of the Hamiltonian matrix (2.352) of a system yields the energy spectrum as well as the state vectors of the system. This procedure, which was worked out by Heisenberg, involves only matrix quantities and matrix eigenvalue equations. This formulation of quantum mechanics is known as *matrix mechanics*.

The starting point of Heisenberg, in his attempt to find a theoretical foundation to Bohr's ideas, was the atomic transition relation, $\nu_{mn} = (E_m - E_n)/h$, which gives the frequencies of the radiation associated with the electron's transition from orbit m to orbit n. The frequencies ν_{mn} can be arranged in a square matrix, where the mn element corresponds to the transition from the mth to the nth quantum state.

We can also construct matrices for other dynamical quantities related to the transition $m \to n$. In this way, every physical quantity is represented by a matrix. For instance, we represent the energy levels by an energy matrix, the position by a position matrix, the momentum by a momentum matrix, the angular momentum by an angular momentum matrix, and so on. In calculating the various physical magnitudes, one has thus to deal with the algebra of matrix quantities. So, within the context of matrix mechanics, one deals with noncommuting quantities, for the product of matrices does not commute. This is an essential feature that distinguishes matrix mechanics from classical mechanics, where all the quantities commute. Take, for instance, the position and momentum quantities. While commuting in classical mechanics, $px = xp$, they do not commute within the context of matrix mechanics; they are related by the commutation relation $[\hat{X},\ \hat{P}_x] = i\hbar$. The same thing applies for the components of angular momentum. We should note that the role played by the commutation relations within the context of matrix mechanics is similar to the role played by Bohr's quantization condition in atomic theory. Heisenberg's matrix mechanics therefore requires the introduction of some mathematical machinery—linear vector spaces, Hilbert space, commutator algebra, and matrix algebra—that is entirely different from the mathematical machinery of classical mechanics. Here lies the justification for having devoted a somewhat lengthy section, Section 2.5, to study the matrix representation of quantum mechanics.

2.7.2 Wave Mechanics

Representing the formalism of quantum mechanics in a *continuous* basis yields an eigenvalue problem not in the form of a matrix equation, as in Heisenberg's formulation, but in the form of a *differential equation*. The representation of the eigenvalue equation (2.351) in the *position* space yields

$$\langle \vec{r} \mid \hat{H} \mid \psi \rangle = E \langle \vec{r} \mid \psi \rangle. \tag{2.353}$$

As shown in (2.315), the Hamiltonian is given in the position representation by $-\hbar^2\nabla^2/(2m) + \hat{V}(\vec{r})$, so we can rewrite (2.353) in a more familiar form:

$$\boxed{-\frac{\hbar^2}{2m}\nabla^2\psi(\vec{r}) + \hat{V}(\vec{r})\psi(\vec{r}) = E\psi(\vec{r}),} \tag{2.354}$$

where $\langle\vec{r} \mid \psi\rangle = \psi(\vec{r})$ is the *wave function* of the system. This differential equation is known as the *Schrödinger equation* (its origin will be discussed in Chapter 3). Its solutions yield the energy spectrum of the system as well as its wave function. This formulation of quantum mechanics in *the position representation* is called *wave mechanics*.

Unlike Heisenberg, Schödinger took an entirely different starting point in his quest to find a theoretical justification for Bohr's ideas. He started from the de Broglie particle–wave hypothesis and extended it to the electrons orbiting around the nucleus. Schrödinger aimed at finding an equation that describes the motion of the electron within an atom. Here the focus is on the *wave* aspect of the electron. We can show, as we did in Chapter 1, that the Bohr quantization condition, $L = n\hbar$, is equivalent to the de Broglie relation, $\lambda = 2\pi\hbar/p$. To establish this connection, we need simply to make three assumptions: (a) the wavelength of the wave associated with the orbiting electron is connected to the electron's linear momentum p by $\lambda = 2\pi\hbar/p$, (b) the electron's orbit is circular, and (c) the circumference of the electron's orbit is an integer multiple of the electron's wavelength, i.e., $2\pi r = n\lambda$. This leads at once to $2\pi r = n \times (2\pi\hbar/p)$ or $n\hbar = rp \equiv L$. This means that, for every orbit, there is only one wavelength (or one wave) associated with the electron while revolving in that orbit. This wave can be described by means of a *wave function*. So Bohr's quantization condition implies, in essence, a uniqueness of the wave function for each orbit of the electron. In Chapter 3 we will show how Schrödinger obtained his differential equation (2.354) to describe the motion of an electron in an atom.

2.8 Concluding Remarks

Historically, the matrix formulation of quantum mechanics was worked out by Heisenberg shortly before Schrödinger introduced his wave theory. The equivalence between the matrix and wave formulations was proved a few years later by using the theory of unitary transformations. Different in form, yet identical in contents, wave mechanics and matrix mechanics achieve the same goal: finding the energy spectrum and the states of quantum systems.

The matrix formulation has the advantage of greater (formal) generality, yet it suffers from a number of disadvantages. On the conceptual side, it offers no visual idea about the structure of the atom; it is less intuitive than wave mechanics. On the technical side, it is difficult to use in some problems of relative ease such as finding the stationary states of atoms. Matrix mechanics, however, becomes powerful and practical in solving problems such as the harmonic oscillator or in treating the formalism of angular momentum.

But most of the efforts of quantum mechanics focus on solving the Schrödinger equation, not the Heisenberg matrix eigenvalue problem. So in the rest of this text we deal mostly with wave mechanics. Matrix mechanics is used only in a few problems, such as the harmonic oscillator, where it is more suitable than Schrödinger's wave mechanics.

In wave mechanics we need only to specify the potential in which the particle moves; the Schrödinger equation takes care of the rest. That is, knowing $\hat{V}(\vec{r})$, we can in principle solve equation (2.354) to obtain the various energy levels of the particle and their corresponding wave

functions. The complexity we encounter in solving the differential equation depends entirely on the form of the potential; the simpler the potential the easier the solution. Exact solutions of the Schrödinger equation are possible only for a few idealized systems; we deal with such systems in Chapters 4 and 6. However, exact solutions are generally not possible, for real systems do not yield themselves to exact solutions. In such cases one has to resort to approximate solutions. We deal with such approximate treatments in Chapters 9 and 10; Chapter 9 deals with time-independent potentials and Chapter 10 with time-dependent potentials.

Before embarking on the applications of the Schrödinger equation, we need first to lay down the theoretical foundations of quantum mechanics. We take up this task in Chapter 3, where we deal with the postulates of the theory as well as their implications; the postulates are the bedrock on which the theory is built.

2.9 Solved Problems

Problem 2.1
Consider the states $|\ \psi\rangle = 9i\ |\ \phi_1\rangle + 2\ |\ \phi_2\rangle$ and $|\ \chi\rangle = -\frac{i}{\sqrt{2}}\ |\ \phi_1\rangle + \frac{1}{\sqrt{2}}\ |\ \phi_2\rangle$, where the two vectors $|\ \phi_1\rangle$ and $|\ \phi_2\rangle$ form a complete and orthonormal basis.

(a) Calculate the operators $|\ \psi\rangle\langle\chi\ |$ and $|\ \chi\rangle\langle\psi\ |$. Are they equal?

(b) Find the Hermitian conjugates of $|\ \psi\rangle$, $|\ \chi\rangle$, $|\ \psi\rangle\langle\chi\ |$, and $|\ \chi\rangle\langle\psi\ |$.

(c) Calculate $\mathrm{Tr}(|\ \psi\rangle\langle\chi\ |)$ and $\mathrm{Tr}(|\ \chi\rangle\langle\psi\ |)$. Are they equal?

(d) Calculate $|\ \psi\rangle\langle\psi\ |$ and $|\ \chi\rangle\langle\chi\ |$ and the traces $\mathrm{Tr}(|\ \psi\rangle\langle\psi\ |)$ and $\mathrm{Tr}(|\ \chi\rangle\langle\chi\ |)$. Are they projection operators?

Solution

(a) The bras corresponding to $|\ \psi\rangle = 9i\ |\ \phi_1\rangle + 2\ |\ \phi_2\rangle$ and $|\ \chi\rangle = -i\ |\ \phi_1\rangle/\sqrt{2} + |\ \phi_2\rangle/\sqrt{2}$ are given by $\langle\psi\ | = -9i\langle\phi_1\ | + 2\langle\phi_2\ |$ and $\langle\chi\ | = \frac{i}{\sqrt{2}}\langle\phi_1\ | + \frac{1}{\sqrt{2}}\langle\phi_2\ |$, respectively. Hence we have

$$\begin{aligned} |\ \psi\rangle\langle\chi\ | &= \frac{1}{\sqrt{2}}\,(9i\ |\ \phi_1\rangle + 2\ |\ \phi_2\rangle)\,(i\langle\phi_1\ | + \langle\phi_2\ |) \\ &= \frac{1}{\sqrt{2}}\,(-9\ |\ \phi_1\rangle\langle\phi_1\ | + 9i\ |\ \phi_1\rangle\langle\phi_2\ | + 2i\ |\ \phi_2\rangle\langle\phi_1\ | + 2\ |\ \phi_2\rangle\langle\phi_2\ |)\,, \end{aligned} \tag{2.355}$$

$$|\ \chi\rangle\langle\psi\ | = \frac{1}{\sqrt{2}}\,(-9\ |\ \phi_1\rangle\langle\phi_1\ | - 2i\ |\ \phi_1\rangle\langle\phi_2\ | - 9i\ |\ \phi_2\rangle\langle\phi_1\ | + 2\ |\ \phi_2\rangle\langle\phi_2\ |)\,. \tag{2.356}$$

As expected, $|\ \psi\rangle\langle\chi\ |$ and $|\ \chi\rangle\langle\psi\ |$ are not equal; they would be equal only if the states $|\ \psi\rangle$ and $|\ \chi\rangle$ were proportional and the proportionality constant real.

(b) To find the Hermitian conjugates of $|\ \psi\rangle$, $|\ \chi\rangle$, $|\ \psi\rangle\langle\chi\ |$, and $|\ \chi\rangle\langle\psi\ |$, we need simply to replace the factors with their respective complex conjugates, the bras with kets, and the kets with bras:

$$|\ \psi\rangle^\dagger = \langle\psi\ | = -9i\langle\phi_1\ | + 2\langle\phi_2\ |, \qquad |\ \chi\rangle^\dagger = \langle\chi\ | = \frac{1}{\sqrt{2}}\,(i\langle\phi_1\ | + \langle\phi_2\ |)\,, \tag{2.357}$$